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ECE PARIS BASE DE RÉSEAUX 2025/2026

TP5

Objectif du TP

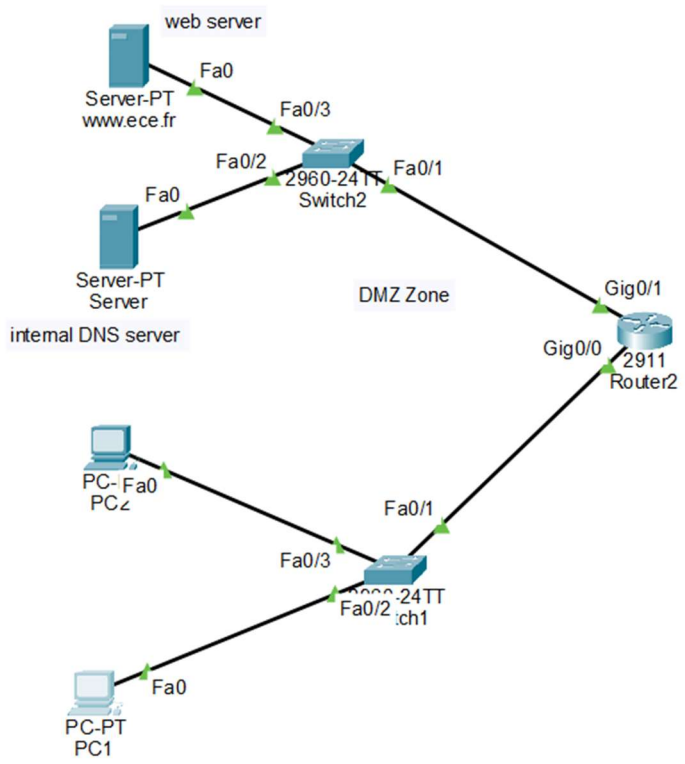
Comprendre le principe d'une architecture client/serveur

Installation et configuration de services réseaux tels que : DHCP, DNS et http/HTTPS

Distinguer les protocoles réseaux de la couche application

Exercice:

En tant qu'administrateur réseau d'une petite entreprise comportant plusieurs serveurs. En utilisant Packet Tracer, vous ajouterez tous les périphériques nécessaires et les connections avec le câblage adéquat. Vous allez vous aider pour cela du schéma ci :



1. **Vérification de la connectivité des périphériques :**

L'objectif dans cette étape est de tester la connectivité entre les périphériques directement connectés.

- A ce stade, PC1 peut-il pinguer les interfaces les serveurs (WEB et DNS) ?

OUI, le PC1 arrive a pinguer les serveurs.

WEB server PC1 → Web Server

```
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time=8ms TTL=127
Reply from 192.168.2.2: bytes=32 time=8ms TTL=127
Reply from 192.168.2.2: bytes=32 time=8ms TTL=127
Reply from 192.168.2.2: bytes=32 time=8ms TTL=127

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 8ms, Maximum = 8ms, Average = 8ms

C:\>
```

DNS Server PC1 → DNS Server

```
C:\>ping 192.168.2.3

Pinging 192.168.2.3 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.3: bytes=32 time=8ms TTL=127
Reply from 192.168.2.3: bytes=32 time=8ms TTL=127
Reply from 192.168.2.3: bytes=32 time=8ms TTL=127

Ping statistics for 192.168.2.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 8ms, Maximum = 8ms, Average = 8ms

C:\>
```

- Testez toutes les liaisons ?

PC1 → Routeur LAN (`192.168.1.1`)

```
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time=4ms TTL=255
Reply from 192.168.1.1: bytes=32 time=4ms TTL=255
Reply from 192.168.1.1: bytes=32 time=4ms TTL=255
Reply from 192.168.1.1: bytes=32 time=4ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 4ms, Maximum = 4ms, Average = 4ms

C:\>
```

PC1 → Routeur DMZ (`192.168.2.1`)

```
C:\>ping 192.168.2.1

Pinging 192.168.2.1 with 32 bytes of data:

Reply from 192.168.2.1: bytes=32 time=4ms TTL=255
Reply from 192.168.2.1: bytes=32 time=4ms TTL=255
Reply from 192.168.2.1: bytes=32 time=4ms TTL=255
Reply from 192.168.2.1: bytes=32 time=4ms TTL=255

Ping statistics for 192.168.2.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 4ms, Maximum = 4ms, Average = 4ms
```

DNS Server → Web Server

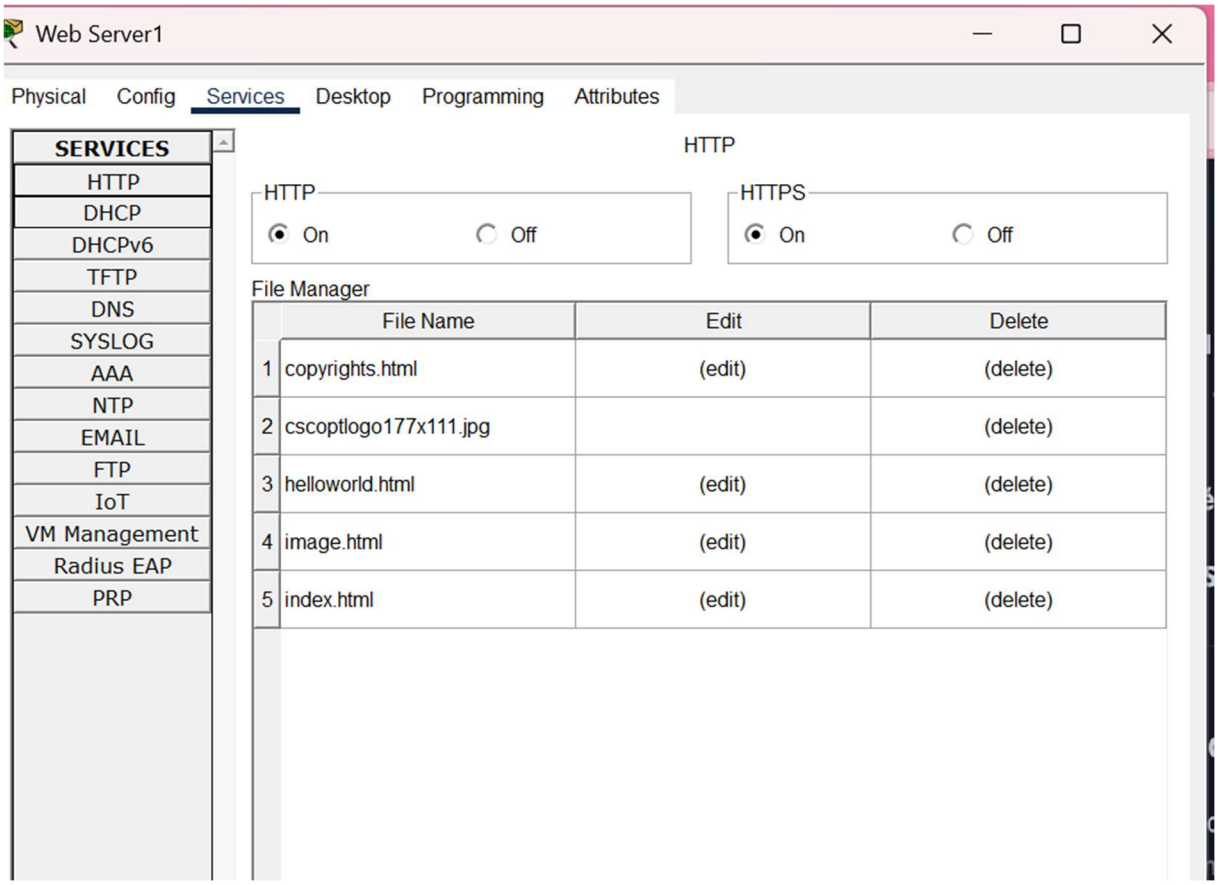
```
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time=8ms TTL=128
Reply from 192.168.2.2: bytes=32 time=4ms TTL=128
Reply from 192.168.2.2: bytes=32 time=4ms TTL=128
Reply from 192.168.2.2: bytes=32 time=4ms TTL=128

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 4ms, Maximum = 8ms, Average = 5ms
```

DNS Server → Routeur DMZ

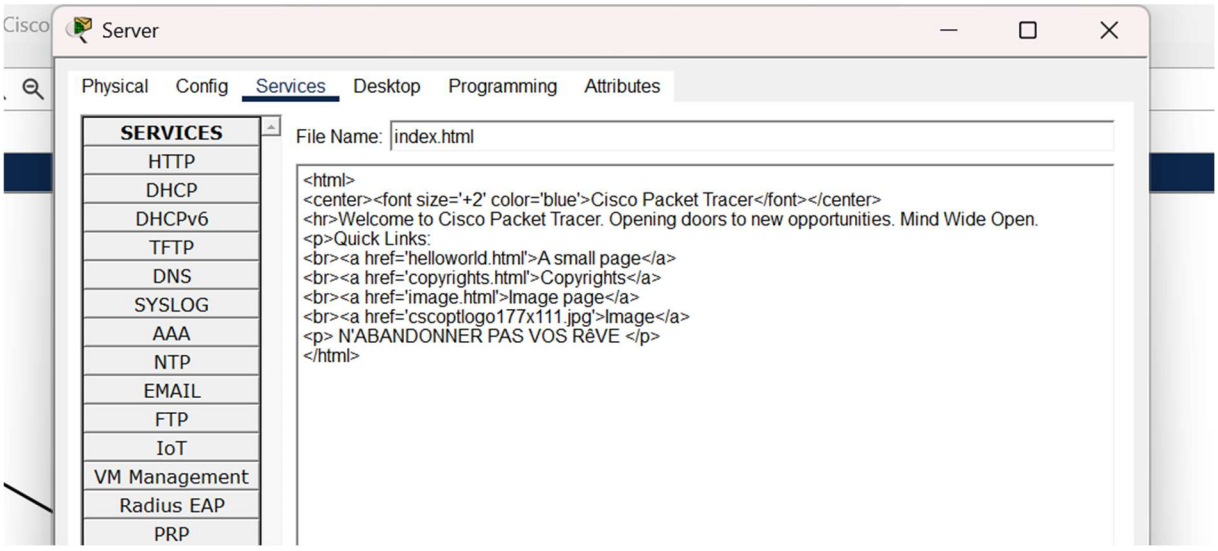
```
C:\>ping 192.168.2.1

Pinging 192.168.2.1 with 32 bytes of data:

Reply from 192.168.2.1: bytes=32 time=4ms TTL=255
Reply from 192.168.2.1: bytes=32 time=4ms TTL=255
Reply from 192.168.2.1: bytes=32 time=4ms TTL=255
Reply from 192.168.2.1: bytes=32 time=4ms TTL=255

Ping statistics for 192.168.2.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 4ms, Maximum = 4ms, Average = 4ms
```

2-Serveur HTTP et serveur DNS Sur DMZ,

a) vérifier que le serveur web est actif.

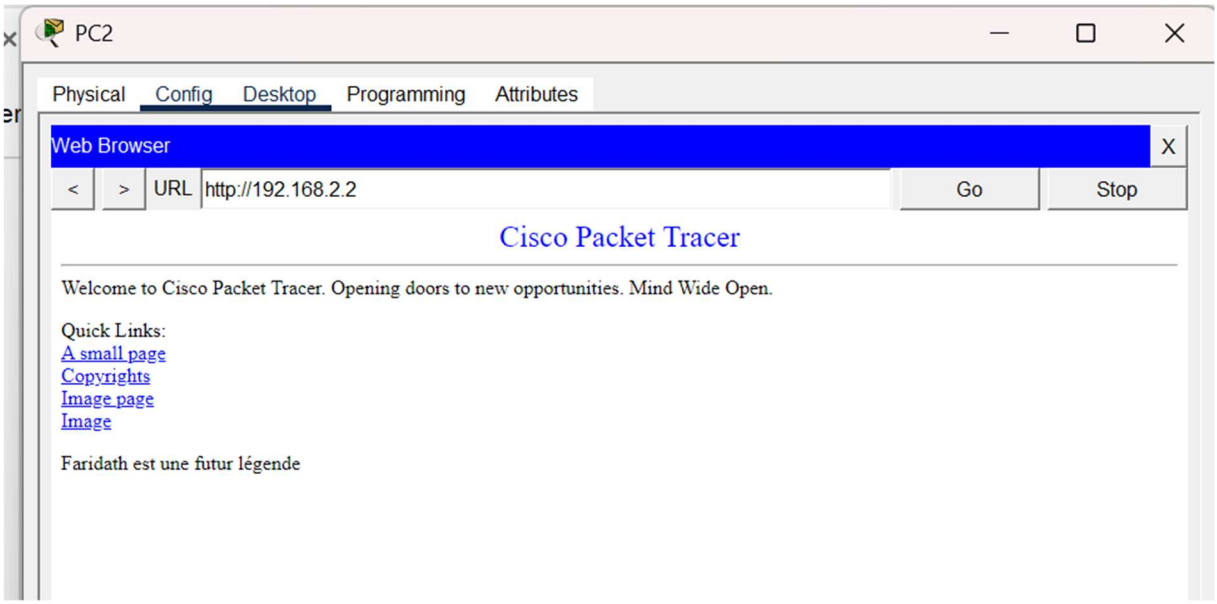


b) Editer les pages HTML d'accueils par défaut pour personnaliser le contenu

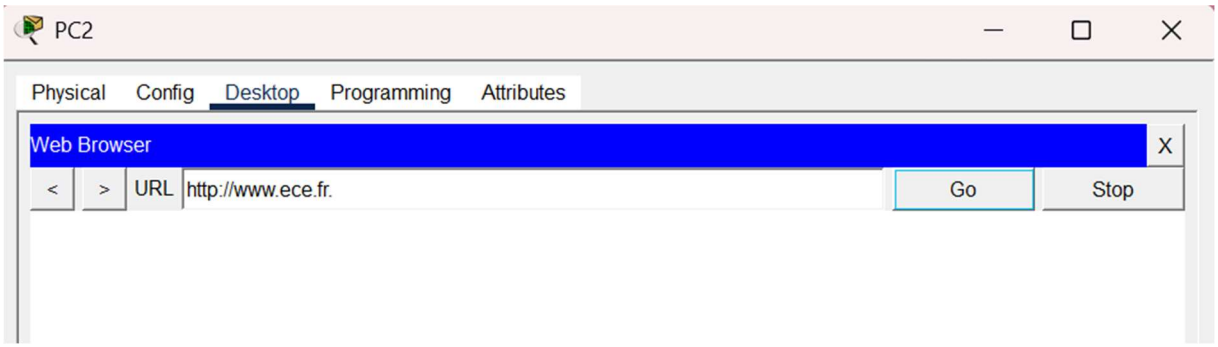


Page HTML modifiée avec succès.

c) Sur LAN, lancer un browser web et accéder à http ://192.168.2.2.



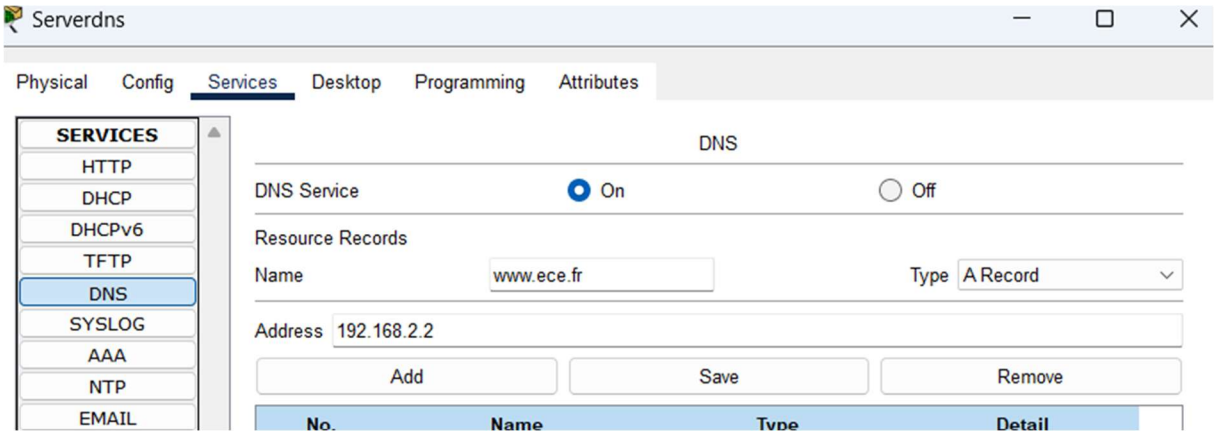
d) Changer le nom du serveur webserver par www.ece.fr et lancer un browser web sur le LAN et accéder à http://www.ece.fr.



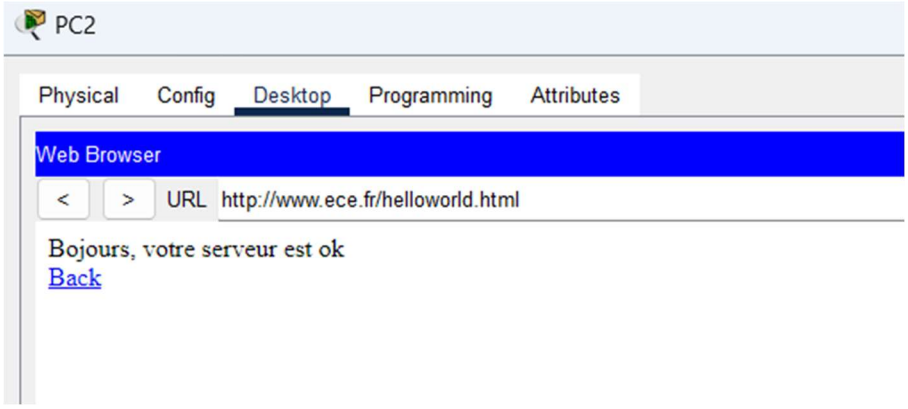
Pourquoi le browser affiche cette réponse ?

Parce que le DNS n'est pas encore configuré. Le PC ne sait pas que www.ece.fr = 192.168.2.2.

e) Configurez le service DNS sur le deuxième serveur Internal_DNS_Serveur afin que les ordinateurs puissent accéder à la page web http://www.ece.fr en tapant une URL plutôt qu'en tapant son adresse IP.

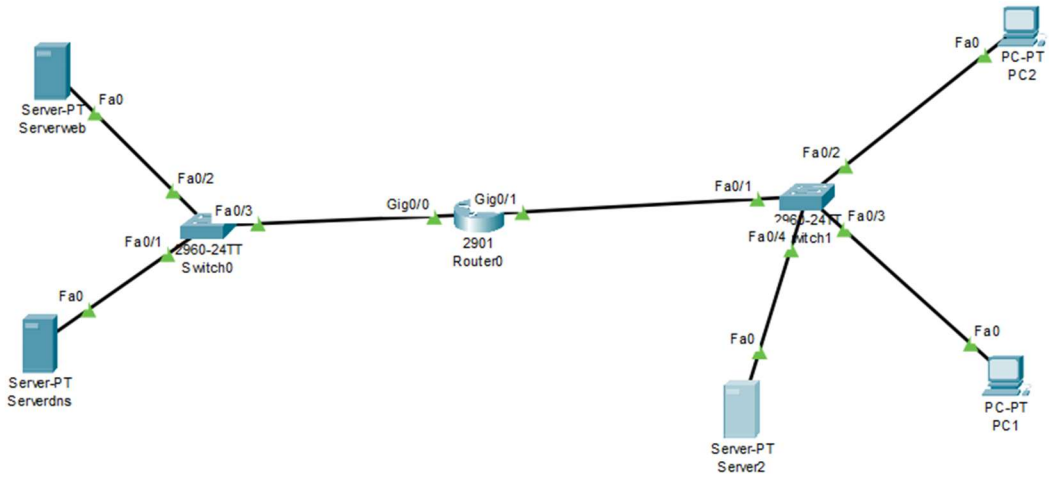


f. Tester et valider votre solution.

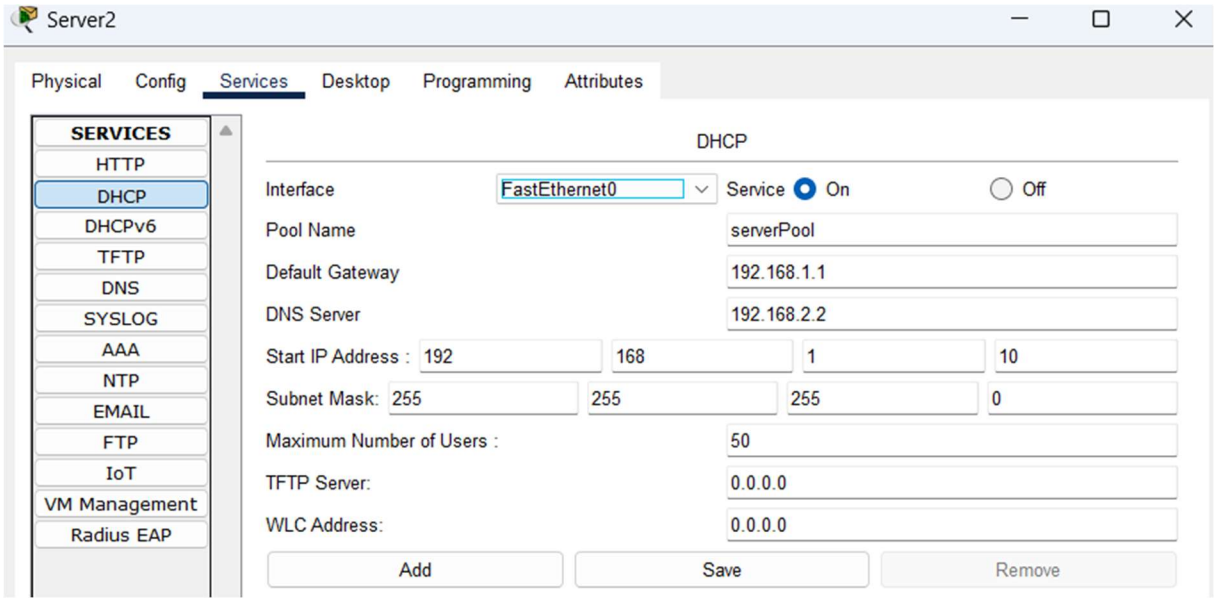


3-Serveur DHCP

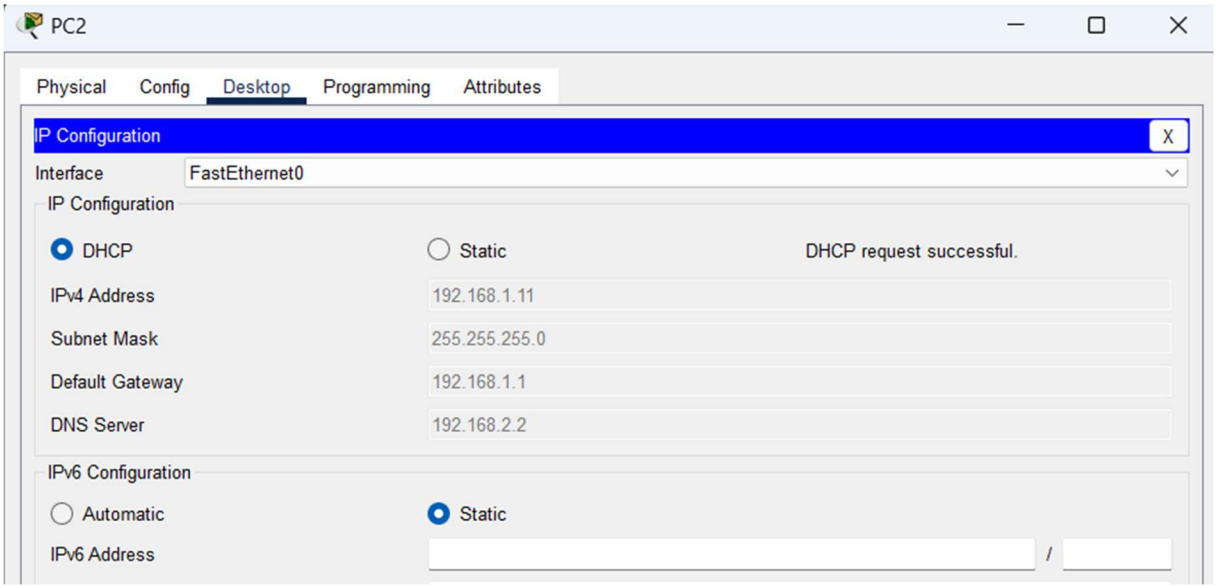
a. Sur LAN, ajouter un serveur pour le déploiement du service DHCP (Dynamic Host Configuration Protocol) qui doit être configurer en statique (adresse IP statique, Passerelle par défaut, DNS).

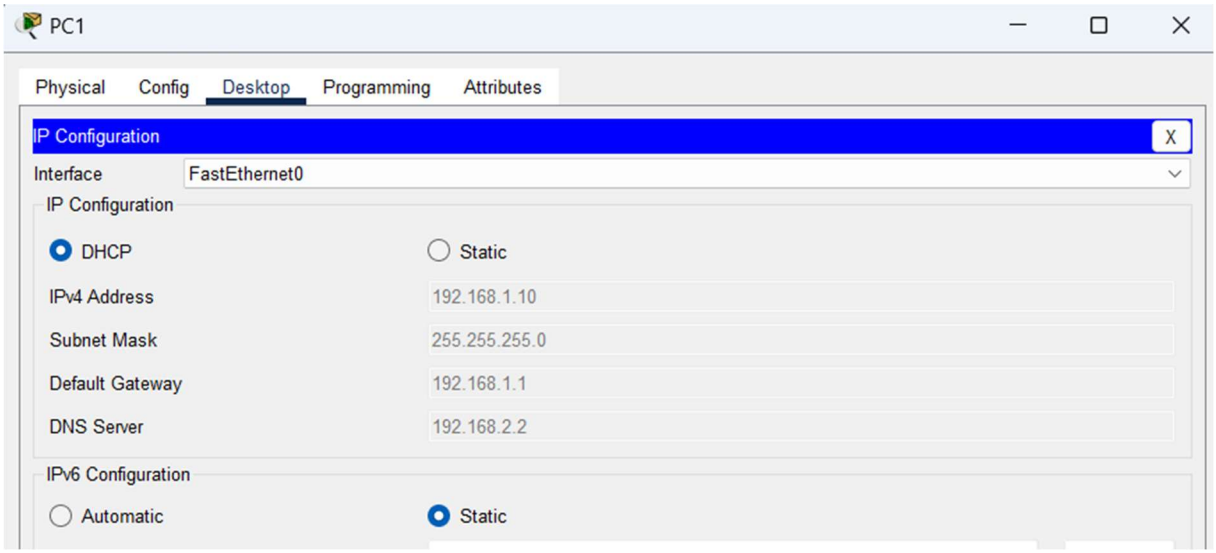


b. Activer le service DHCP sur ce serveur et faites le paramétrage nécessaire pour attribuer dynamiquement les paramètres réseaux aux PCs configurés en dynamique sur ce réseau (plage d'adresses, passerelle par défaut, DNS, durée des bauds,) .



c. Changer la configuration réseau sur les PCs en les mettant en dynamique et non pas en statique.





d. Tester et valider votre solution.

TEST ROUTEUR

```
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time=1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

TEST DNS

```
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time<1ms TTL=127
Reply from 192.168.2.2: bytes=32 time=1ms TTL=127
Reply from 192.168.2.2: bytes=32 time<1ms TTL=127
Reply from 192.168.2.2: bytes=32 time=1ms TTL=127

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

TEST WEB

```
C:\>ping www.ece.fr

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time<1ms TTL=127
Reply from 192.168.2.2: bytes=32 time<1ms TTL=127
Reply from 192.168.2.2: bytes=32 time<1ms TTL=127
Reply from 192.168.2.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```